Draft: game-built quadratic forms in the nimber Clifford backend

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Abstract

This is a draft experimental note about the part of `pleroma` that seems mathematically worth isolating. Conway games under disjunctive sum do not form a scalar ring, so the project does not construct "Clifford algebras over all games". Instead it builds Clifford algebras over field-like game-value worlds, with emphasis on the characteristic-2 nimber backend. The main point is that the char-2 engine keeps the quadratic data $q_i = e_i^2$ and the polar data $b_{ij} = e_ie_j + e_je_i$ independent, then computes trace-Arf invariants of quadratic forms over finite nimber subfields.

The experimental bridge is this: the Gold trace forms

$$Q_a(x) = \mathrm{Tr}(x\, x^{2^a})$$

are standard quadratic forms over fields of characteristic 2, but in the nimber setting they are also composites of combinatorial-game operations: nim-product via Turning Corners, Frobenius squaring via the diagonal product, and XOR via disjunctive sum. Their Arf invariant has the classical zero-count interpretation. Therefore, if a natural game had P-positions exactly $\{x : Q_a(x) = 0\}$, the Arf invariant would be the sign of the second-player win-bias. The note does not claim such a game has been found. It records the construction, the validation checks, and the current obstructions.

1 Scope

A short partizan game is built recursively as $\{L \mid R\}$, and games form a partially ordered abelian group under disjunctive sum [4, 3]. They do not form a ring: Conway multiplication is defined on the number subclass, not on arbitrary games. Since a Clifford algebra needs a commutative scalar ring, the phrase "Clifford over games" has to be interpreted through scalar subworlds:

world	algebraic structure	role here
No	real-closed field, char 0	ordinary real Clifford theory with surreal scalars
No$[i]$	algebraically closed field, char 0	ordinary complex Clifford theory
finite nimber subfields	finite fields of char 2	characteristic-2 Clifford and Arf experiments

The implemented `Nimber` scalar is finite: `u128` realizes $\mathbb{F}_{2^{128}}$, containing the nimber subfields $\mathbb{F}_{2^m}$ for $m = 1, 2, 4, \ldots, 128$. It is not the whole proper-class field On_2.

2 The char-2 Clifford datum

In characteristic 2, a quadratic form is not determined by its polar form. For basis vectors the Clifford relations are

$$e_i^2 = q_i, \qquad e_ie_j + e_je_i = b_{ij}.$$

The polar form is alternating, so $b_{ii} = 0$, but q_i can be nonzero. A faithful characteristic-2 implementation must store q and b separately. This is the main implementation discipline behind the nimber backend.

For a nonsingular quadratic form over $\mathbb{F}_2$ with symplectic basis (a_k, b_k), the Arf invariant is

$$\mathrm{Arf}(Q) = \sum_k Q(a_k)Q(b_k) \in \mathbb{F}_2.$$

It classifies nonsingular binary quadratic spaces at fixed dimension. Over a finite characteristic-2 field, the corresponding value lies in $F/\wp(F)$, with $\wp(y) = y^2 + y$; for finite fields this quotient is read by the field trace to $\mathbb{F}_2$.

Remark 1. Ovsienko [8] identifies a precise bridge between Dickson's classification of quadratic forms over $\mathbb{F}_2$ and the classical classification of real Clifford algebras. We use that paper as motivation for the Arf/Clifford connection, not as a blanket classification theorem for all Clifford algebras over finite nimber fields. In the code, rank and radical data are reported alongside Arf, and degenerate forms are not collapsed to a single Arf bit.

The small finite-field check used throughout is:

$$q = (*2, *2),\ b_{01} = *1 \quad \text{has Arf } 1,$$

whereas

$$q = (*2, *3),\ b_{01} = *1 \quad \text{has Arf } 0.$$

The polar entry is part of the form.

3 Gold forms from game operations

Nim-addition is XOR, and XOR is the disjunctive sum of impartial game values. Nim-multiplication is also game-theoretic: Conway's Turning-Corners product has the mex recurrence

$$x \otimes y = \mathrm{mex}\left\{(i \otimes y) \oplus (x \otimes j) \oplus (i \otimes j) : 0 \le i < x,\ 0 \le j < y\right\}. \tag{1}$$

The repo implements this recurrence directly for small arguments and checks it against the fast algebraic nim product.

Thus the following operations are game-realizable at the level of nimber values:

$$x \oplus y, \qquad x \otimes y, \qquad x \mapsto x^2 = x \otimes x, \qquad \mathrm{Tr}(x) = x + x^2 + \cdots + x^{2^{m-1}}.$$

For $F = \mathbb{F}_{2^m}$ and $a \ge 1$, define the Gold trace form

$$Q_a(x) = \mathrm{Tr}\big(x\, x^{2^a}\big) = \mathrm{Tr}\big(x^{1+2^a}\big).$$

Since Frobenius $x \mapsto x^{2^a}$ is additive, Q_a is a quadratic form over $\mathbb{F}_2$. In the nimber backend, it is also a composite of the game operations above.

Observation 1. *The Gold forms are not merely defined on a field of game values. In the finite nimber fields tested here, they are built from nim/game operations: Turning Corners for multiplication, diagonal Turning Corners for Frobenius squaring, and disjunctive sum for XOR and trace.*

This observation is modest but useful: it gives a concrete quadratic form at the interface between impartial-game arithmetic and characteristic-2 quadratic-form theory.

3.1 Rank check

The polar rank of Gold forms is known. In the tested regime $m = 2^k$ and $1 \le a \le k$, the general formula

$$\operatorname{rank} B_{Q_a} = m - \gcd(2a, m)$$

specializes to $m - 2\gcd(a, m)$ because $m/\gcd(a, m)$ is even. The classifier reproduces the general formula in all tested cases. Representative rows are shown in Table 1.

field	a	Q	Arf	type	rank	formula
$\mathbb{F}_{2^4}$	1	$\mathrm{Tr}(x^3)$	1	O^-	2	2
$\mathbb{F}_{2^8}$	1	$\mathrm{Tr}(x^3)$	1	O^-	6	6
$\mathbb{F}_{2^8}$	2	$\mathrm{Tr}(x^5)$	1	O^-	4	4
$\mathbb{F}_{2^{16}}$	1	$\mathrm{Tr}(x^3)$	1	O^-	14	14
$\mathbb{F}_{2^{16}}$	4	$\mathrm{Tr}(x^{17})$	1	O^-	8	8
$\mathbb{F}_{2^{32}}$	1	$\mathrm{Tr}(x^3)$	1	O^-	30	30

Table 1: Representative output of `experiments/trace_form_arf.py`. The script checks all 15 cases with $m = 2, 4, 8, 16, 32$ and $1 \le a \le \log_2 m$.

Most of these forms have a nontrivial radical. The statement is therefore not "nondegenerate Gold form" but "positive-rank quadratic form with a nonsingular core whose Arf value is computed after symplectic reduction."

4 A broader trace family

The coefficient-1 Gold form is only one trace monomial. A standard trace description of the quadratic part of Boolean functions over $\mathbb{F}_{2^m}$ uses terms

$$\mathrm{Tr}(c_i x^{1+2^i})$$

plus affine terms and, in even degree, a middle half-trace term. The monomials displayed here are still game-operation-built in the nimber backend: $c_i x^{1+2^i} = c_i \otimes x \otimes x^{2^i}$, followed by trace/XOR.

The current implementation does not claim to package every Boolean quadratic function. The probe `experiments/gold_family_survey.py` deliberately omits the middle term and affine bookkeeping. What it does check, exhaustively over $\mathbb{F}_{256}$, is that scaled Gold components

$$Q_{\lambda,a}(x) = \mathrm{Tr}(\lambda x^{1+2^a})$$

already reach nondegenerate (bent) forms for many λ. For APN Gold exponents in that field, the observed count matches the classical $2(2^m - 1)/3$ bent-component count. This matters for the open game question because a bent form has trivial radical: the loopy radical route collapses to $\{0\}$, and the frame-blind symplectic no-go applies without a degenerate layer to hide in.

5 Arf as conditional win-bias

The relevant counting theorem is classical. If Q is nonsingular on $\mathbb{F}_2^{2r}$, then

$$\#\{x : Q(x) = 0\} = 2^{2r-1} + (-1)^{\mathrm{Arf}(Q)} 2^{r-1}.$$

For degenerate forms, the code uses the standard radical-adjusted count: an anisotropic radical balances the values, while an isotropic radical scales the bias of the nonsingular core.

The game interpretation is conditional:

> If a game has P-positions exactly $\{x : Q(x) = 0\}$, then the Arf invariant is the sign of the bias between second-player wins and first-player wins.

This is not a claim that such a game has been found. It is a target for a game semantics.

field	a	Arf	rank	$\#\{Q = 0\}$	bias
$\mathbb{F}_{2^4}$	1	1	2	4	-4
$\mathbb{F}_{2^8}$	1	1	6	112	-16
$\mathbb{F}_{2^8}$	2	1	4	96	-32
$\mathbb{F}_{2^{16}}$	1	1	14	32512	-256
$\mathbb{F}_{2^{16}}$	4	1	8	30720	-2048

Table 2: Representative zero-counts from `experiments/arf_win_bias.py`. The bias is $\#\{Q = 0\} - 2^{m-1}$.

6 What blocks the game semantics

For a normal-play disjunctive sum of impartial games, the P-condition is linear in Grundy coordinates:

$$g_1 \oplus \cdots \oplus g_n = 0.$$

So normal-play sums give linear P-sets, not genuine quadrics.

For a quadratic form in characteristic 2,

$$Q(u + v) = Q(u) + Q(v) + B(u, v).$$

Thus for $u, v \in \{Q = 0\}$,

$$u + v \in \{Q = 0\} \iff B(u, v) = 0.$$

The polar form is exactly the obstruction to XOR-closure. The first two layers are game-built: XOR is disjunctive sum and B is built from nim products. The missing layer is a play rule that integrates that bilinear data into the quadratic zero set.

6.1 Test benches

The repo contains several probes. They are useful for narrowing the target, but they are not a proof that no natural game exists.

- `fit_f2_quadratic` tests whether a subset of $\mathbb{F}_2^k$ is the zero set of a quadratic polynomial, and whether the result has nonzero polar rank.

- A bounded misere quotient computer tests small quotient P-sets. The octal sweep checked 292 octal codes, heap cutoff at most 4, and found quotient orders $2, 6, 10, 12, 14$, with no elementary abelian 2-group of rank at least 2.

- `experiments/misere_kernel.py` checks the kernel obstruction on the smallest wild misere quotient R_8: the canonical group/kernel shadow is linear, while the genuinely misere P-element lies outside that vector-space part.

- An interactive-kernel solver tests finite move graphs. It confirms that an ad hoc acyclic game can realize any subset, and that a rule referencing Q itself realizes $\{Q = 0\}$ tautologically. The tested B-coupled rules do not produce the Gold zero set.

- `examples/loopy_quadric.rs` tests cyclic move graphs with Draw sets. The symmetric B-coupled rule has Loss-set equal to the radical $R(B)$, so it reproduces the obstruction rather than the Gold zero set.

- `examples/bent_route.rs` repeats route probes on a bent scaled Gold component. In that example, a B plus coordinate-frame rule reaches a quadric of the correct Arf class but not the specific Gold zero set; the naive local-field/spin-flip completion leaves the quadric variety.

Except for the frame-blind proposition below and the quoted misere-kernel structure theorem, these are bounded or example-level negative results.

6.2 Frame-blind no-go

There is, however, a clean obstruction for rules that see only a nondegenerate symplectic form and no frame.

Proposition 1. *Let B be a nondegenerate alternating form on $V = \mathbb{F}_2^{2r}$ with $r \geq 2$. If a finite normal-play game has position set V and a move relation invariant under* $\mathrm{Sp}(B)$*, then its P-set is not a nondegenerate quadric.*

Proof. Each element of $\mathrm{Sp}(B)$ is an automorphism of the move graph, so the retrograde outcome function is $\mathrm{Sp}(B)$-invariant. The P-set is therefore a union of symplectic orbits. For $r \geq 2$, the symplectic group is transitive on $V \setminus \{0\}$ [10], so the invariant subsets are only $\varnothing$, $\{0\}$, $V \setminus \{0\}$, and V. A nondegenerate quadric in dimension at least 4 has $2^{2r-1} \pm 2^{r-1}$ points, which is none of these. □

The hypothesis matters. Coordinate-dependent rules have already broken $\mathrm{Sp}(B)$ symmetry, so this proposition does not explain every negative probe. It explains only genuinely frame-blind B-only rules.

6.3 The diagonal framing

Once a coordinate frame is allowed, a canonical-looking quadratic refinement of B is

$$Q_{\mathrm{frame}}(v) = \sum_{i<j} B(e_i, e_j) v_i v_j.$$

For the Gold polar forms tested in this repo, the Gold form decomposes as

$$Q_a(v) = Q_{\mathrm{frame}}(v) + \ell_{\mathrm{diag}}(v), \qquad \ell_{\mathrm{diag}}(v) = \sum_i Q_a(e_i) v_i.$$

In the tested genuinely quadratic Gold cases up to $\mathbb{F}_{2^{16}}$, Q_{frame} has Arf 0 and the diagonal term changes the form to the Gold Arf-1 class.

Remark 2. This is an experimental pattern for the Gold polar forms tested here, not a theorem about every alternating form plus every coordinate frame. Random nondegenerate alternating forms can give a zero-diagonal frame quadric of Arf 1.

The remaining question is therefore sharper than the original slogan.

Open question. *Is the diagonal framing*

$$q_i = Q_a(e_i) = \mathrm{Tr}(e_i^{1+2^a})$$

itself game-natural, in a way that yields a natural game with P-set $\{x : Q_a(x) = 0\}$*?*

7 Validation status

The claims above are backed by implementation checks, not by a finished proof of the open game statement.

- `cargo test` exercises the scalar arithmetic, Clifford product, Arf computation, quadratic fitting, misere machinery, and related invariant modules.
- `experiments/trace_form_arf.py` checks the Gold rank formula.
- `experiments/gold_form_from_games.py` and `experiments/tartan_bilinear.py` rebuild the Gold form and polar form from literal Turning-Corners products in small fields.
- `experiments/arf_win_bias.py` checks the zero-count formula.
- The `misere_quotient`, `octal_hunt`, `interactive_kernel`, `loopy_quadric`, and `bent_route` examples implement the current route probes.
- `experiments/gold_family_survey.py` checks the scaled-component bent examples discussed above.
- `experiments/misere_kernel.py` checks the misere-kernel reading on R_8.

8 Conclusion

The meaningful content is the construction of game-operation-built characteristic-2 trace forms inside a nimber Clifford backend, together with a precise conditional interpretation of Arf as a P-set bias. The draft does not solve the game-semantics problem. It narrows the current search to a named missing datum: a non-tautological way for play to use the diagonal quadratic framing that turns the game-built polar form into the target quadric.

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